

Joshua Brenneman

Senior Software Engineer · AWS Cloud & Platform Engineering

jbrenn.dev@gmail.com github.com/jabrennem linkedin.com/in/joshuabrennemana

SUMMARY

Senior Software Engineer with a decade of experience designing and building AWS-native platforms for biomedical research, clinical genomics, data engineering, and machine learning. Skilled at translating complex scientific problems into reliable, production-grade cloud systems. Background in linguistics and computational analysis. AWS Certified Developer – Associate.

EXPERIENCE

Senior Software Engineer

Apr 2023 – Present

Nationwide Children's Hospital · Office of Data Sciences, Cloud Solutions Team

- Lead architect of a multi-account research data lakehouse on AWS, managing catalog (Glue), storage (S3), ETL (Step Functions, EMR Serverless), governance (Lake Formation, IAM), and lineage (EventBridge).
- Build APIs via API Gateway, EventBridge, and MCPs for data producers and consumers to integrate with the lakehouse.
- Partner with data scientists on large-scale NLP applications using Apache Iceberg and in-house data lake infrastructure.
- Integrate generative AI into internal tools using Amazon Bedrock and Model Context Protocol.
- Worked with solutions architect to build a multi-account ecosystem using AWS Organizations with 10+ accounts encompassing petabytes of data.
- Built CI/CD pipelines with AWS CodePipeline, CodeBuild, and GitHub Actions for production deployments.
- Migrated the Data Lakehouse to a NIST-compliant environment.

Bioinformatics Software Developer – Research

Oct 2019 – Apr 2023

Nationwide Children's Hospital · Institute for Genomic Medicine, Cloud Solutions Team

- Designed event-driven genomics workflows using Step Functions, Lambda, and AWS Batch to process next-generation sequencing pipelines.
- Contributed to infrastructure for Rapid Genome Sequencing, a protocol serving NICU patients.
- Built and maintained a genomic variant data lake using S3, Glue, Athena, and EMR Serverless.
- Collaborated with bioinformatics scientists to translate research workflows into reliable, cost-effective production systems.
- Built serverless web applications with React, S3/CloudFront, API Gateway, Lambda, and DynamoDB for scientists and directors.

Bioinformatics Software Developer – Clinical

Oct 2017 – Oct 2019

Nationwide Children's Hospital · Institute for Genomic Medicine, Clinical Informatics Team

- Wrote software for production clinical genomics operations and next-generation sequencing pipelines in collaboration with clinical directors.

Big Data Application Developer

Jun 2016 – Oct 2017

CGI · Consulting engagement with PNC Bank

- Maintained and developed ETL pipelines in an on-premise Hadoop and Apache Spark ecosystem.

CAPABILITIES

Platform Engineering: AWS architecture, serverless systems, distributed workflows, event-driven architecture, developer enablement.

Compute & Containers: Lambda, AWS Batch, Docker, production orchestration.

Software Engineering: Python, TypeScript, JavaScript for application architecture and product development.

Cloud & Data: S3, Athena, Apache Iceberg, Glue, EMR, ETL, lakehouse architecture.

Infrastructure & Delivery: CDK, CloudFormation, SAM, GitHub Actions, CodePipeline, CI/CD.

AI Platforms: SageMaker, Bedrock, agentic architecture, MCP, AI-assisted engineering.

EDUCATION

B.A. Linguistics, Minor in Computer Science

CERTIFICATION

AWS Certified Developer – Associate

LANGUAGES

German, Italian